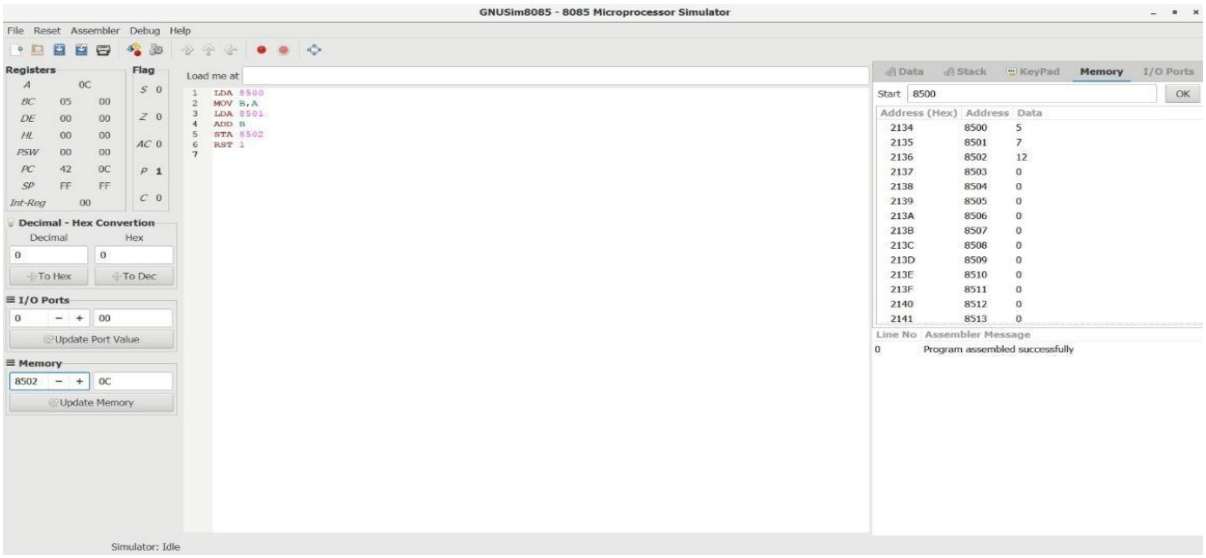
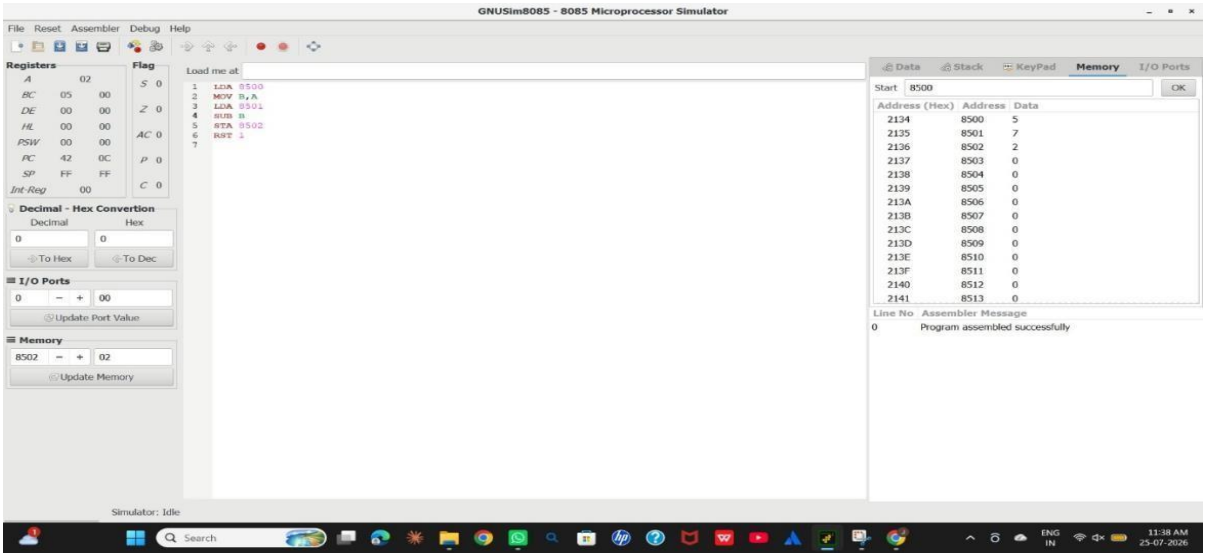
CSA1204 COMPUTER ARCHITECTURE

8085 LAB PROGRAMS

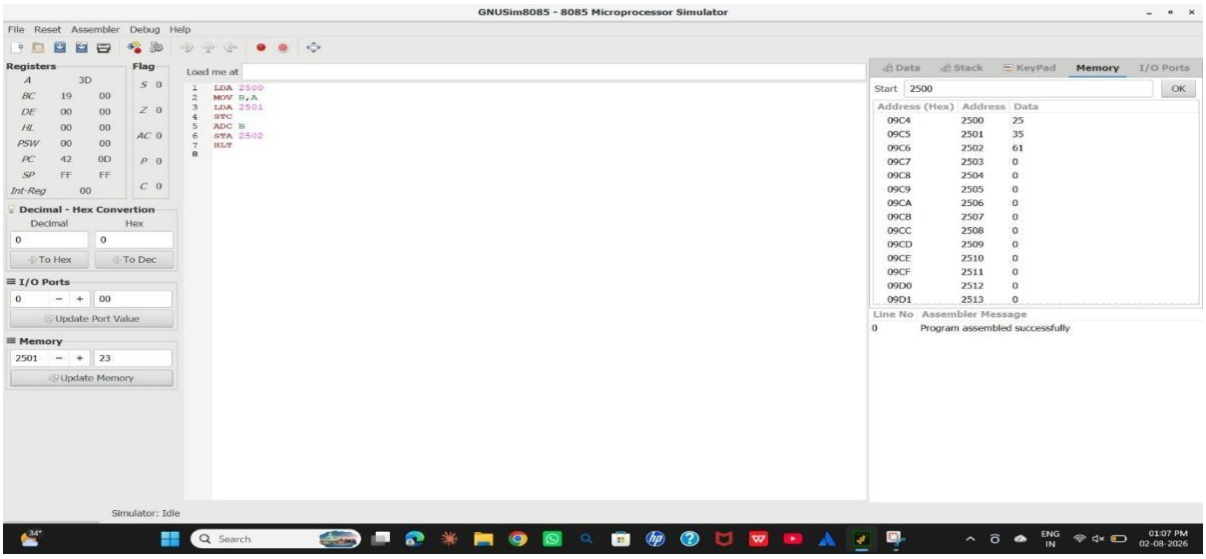
Q1. Addition of 2 Numbers



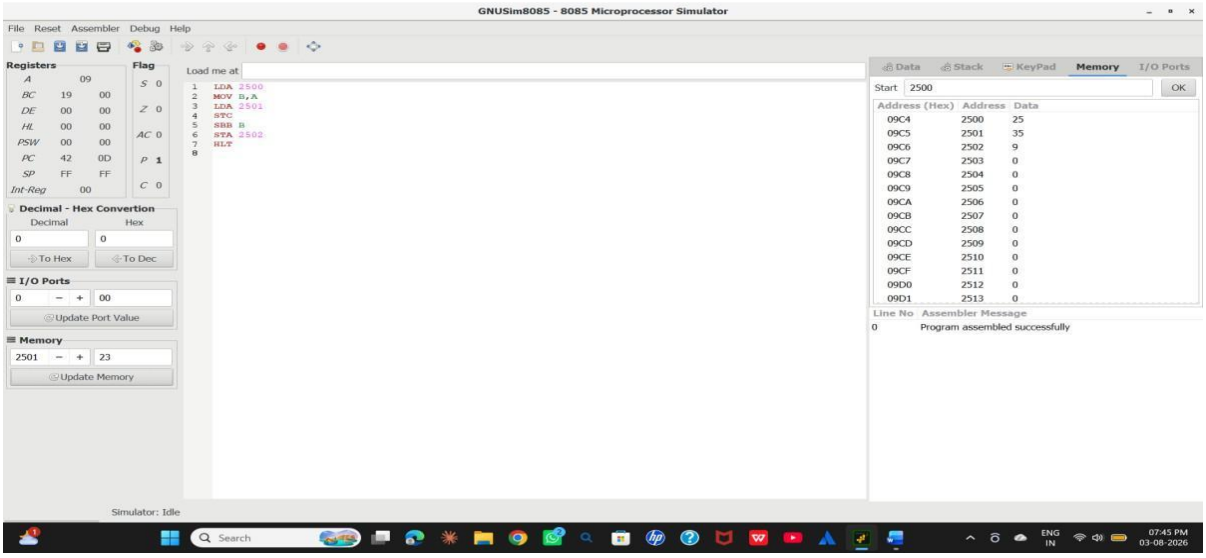
Q2. Subtraction of 2 Numbers



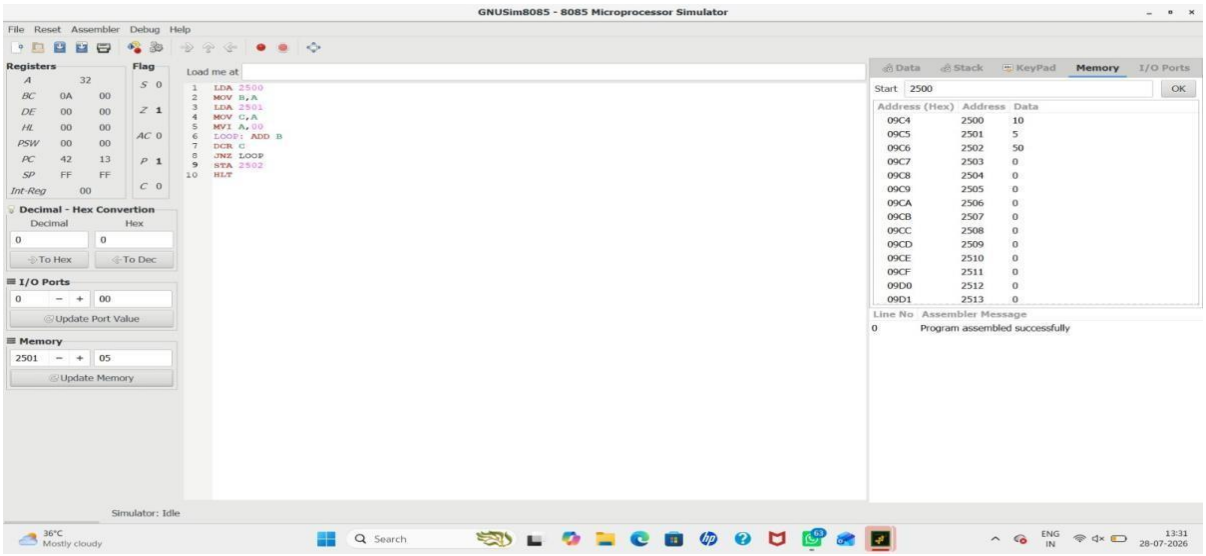
Q3. Addition With Carry (ADC)



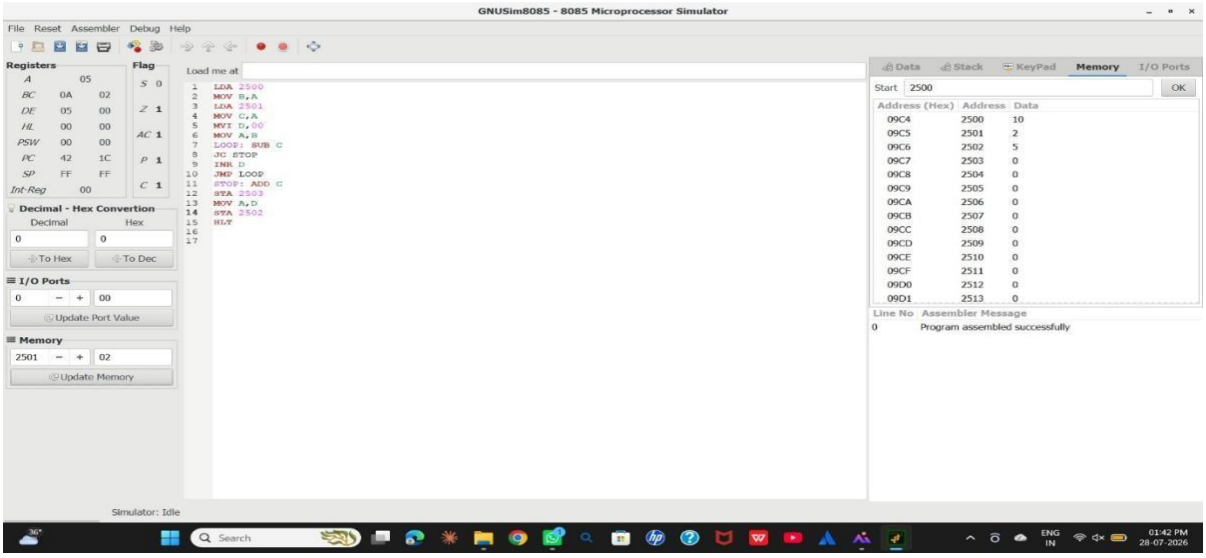
Q4. Subtraction With Borrow (SBB)



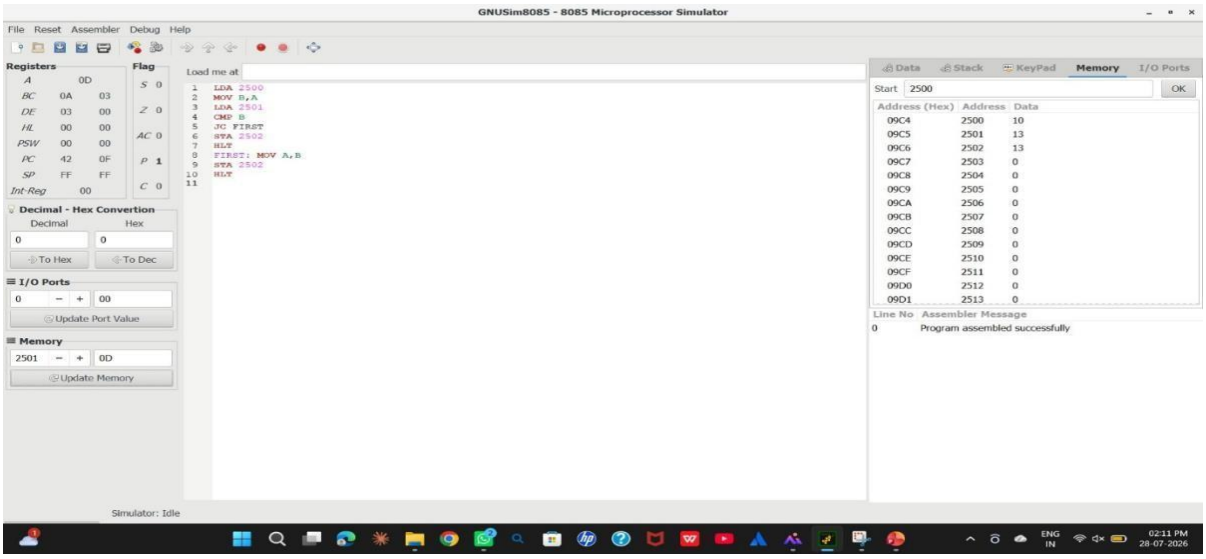
Q5. Multiplication of 2 Numbers



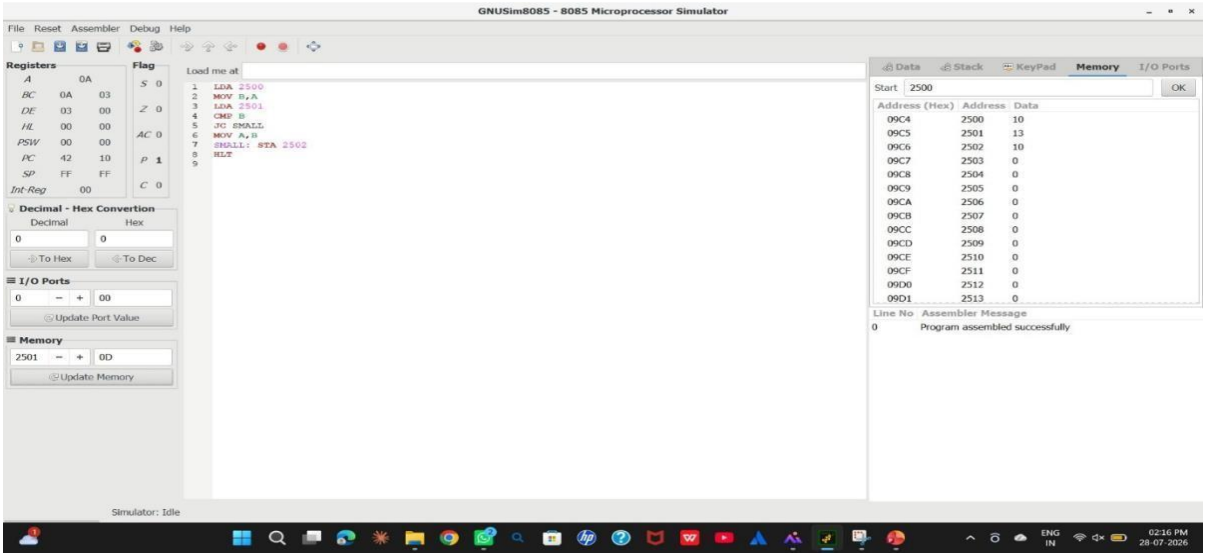
Q6. Division of Two Numbers



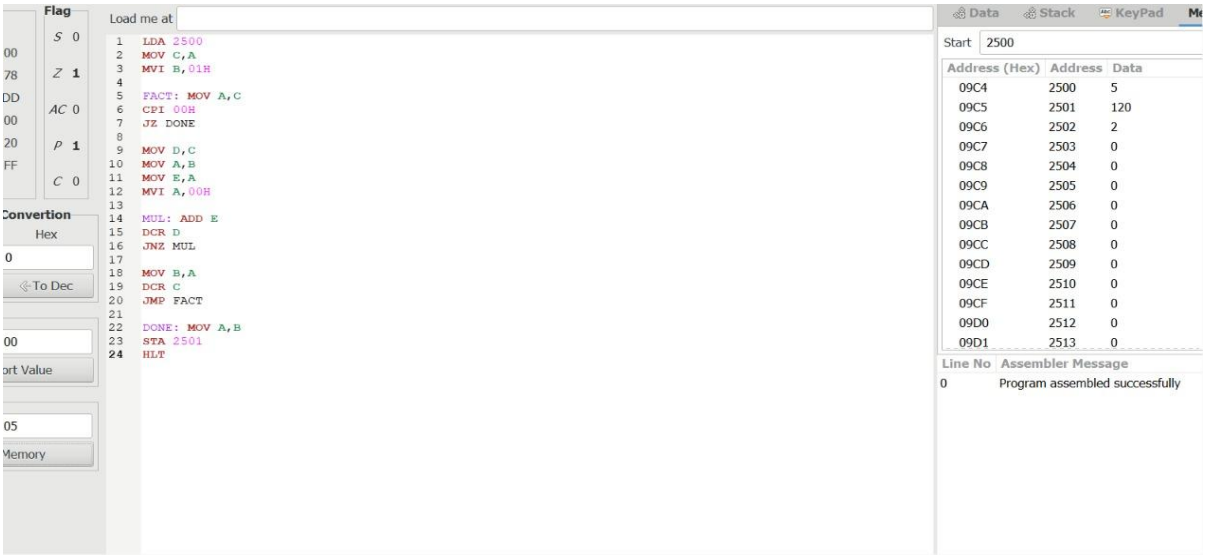
Q7. Greatest of 2 Numbers



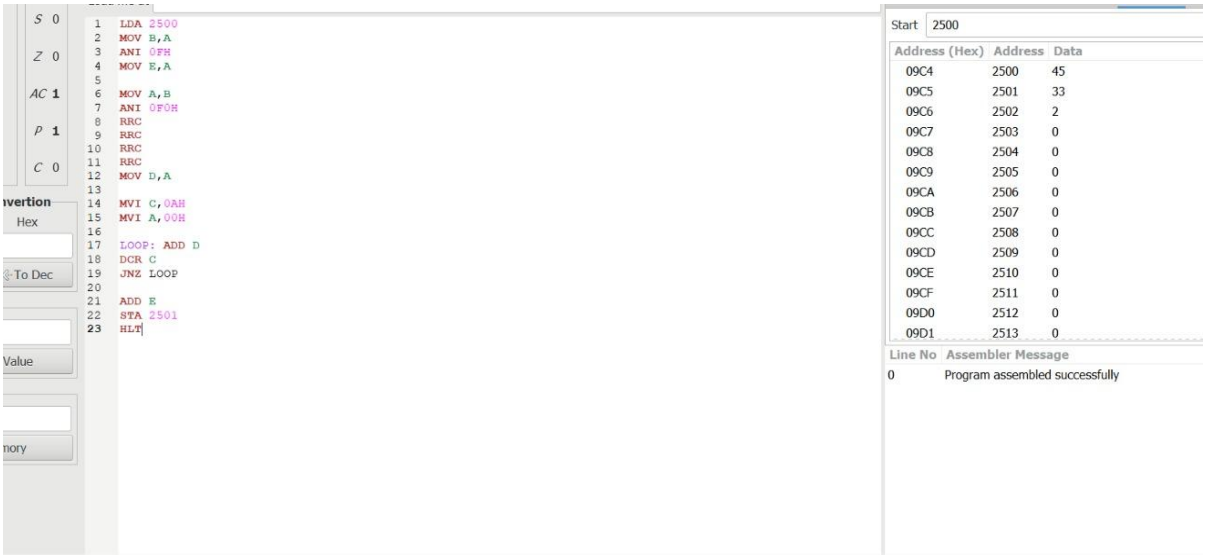
Q8. Smallest of 2 Numbers



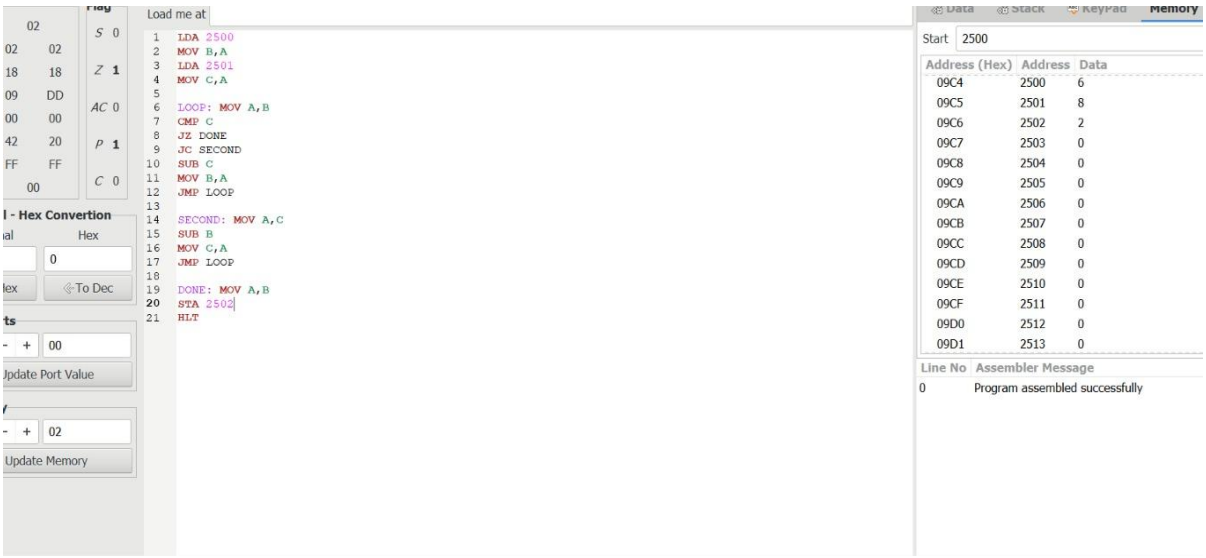
Q9. Factorial of a Number



Q10. Packed BCD to Binary Conversion



Q11. GCD (HCF) of Two Numbers



```
LDA 2500
MOV B,A
LDA 2501
MOV C,A

LOOP: MOV A,B
CMP C
JZ DONE
JC SECOND
SUB C
MOV B,A
JMP LOOP

SECOND: MOV A,C
SUB B
MOV C,A
JMP LOOP

DONE: MOV A,B
STA 2502
HLT
```

Address (Hex)	Address	Data
09C4	2500	6
09C5	2501	8
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No	Assembler Message
0	Program assembled successfully

Q12. LCM of Two Numbers

```
LDA 2500
MOV B,A
LDA 2501
MOV C,A

MOV D,B
MOV E,C

LOOP: MOV A,D
CMP E
JZ DONE
JC ADDD

MOV A,E
ADD C
MOV E,A
JMP LOOP

ADDD: MOV A,D
ADD B
MOV D,A
JMP LOOP

DONE: MOV A,D
STA 2502
HLT
```

Address (Hex)	Address	Data
09C4	2500	6
09C5	2501	8
09C6	2502	24
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No	Assembler Message
0	Program assembled successfully

Q13. Smallest Number in an Array

```
Load me at

1  LXI H,2501
2  MOV B,M
3  MOV A,M
4  MOV D,A
5  DCR B
6
7  LOOP: INX H
8  MOV A,M
9  CMP D
10 JNC SKIP
11 MOV D,A
12
13 SKIP: DCR B
14 JNZ LOOP
15
16 MOV A,D
17 STA 2600
18 HLT
```

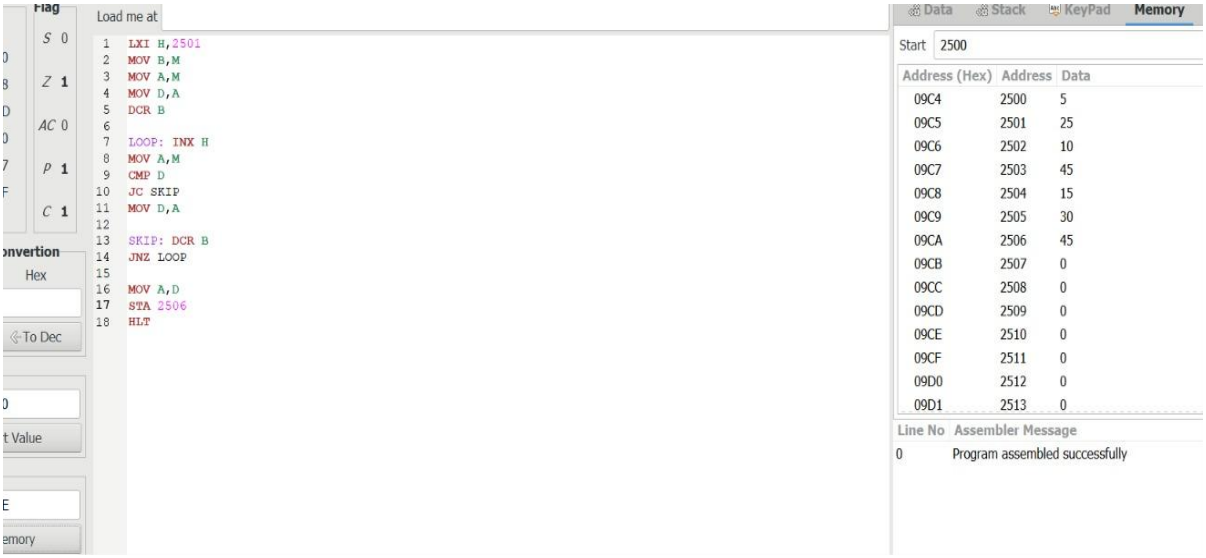
Data | Stack | KeyPad | **Memory**

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	5
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No	Assembler Message
0	Program assembled successfully

Q14. Largest Number in an Array



```
Load me at

1  LXI H,2501
2  MOV B,M
3  MOV A,M
4  MOV D,A
5  DCR B
6
7  LOOP: INX H
8  MOV A,M
9  CMP D
10 JC SKIP
11 MOV D,A
12
13 SKIP: DCR B
14 JNZ LOOP
15
16 MOV A,D
17 STA 2506
18 HLT
```

Flag: S 0, Z 1, AC 0, P 1, C 1

onvertion

Hex

To Dec

t Value

emory

Data | Stack | KeyPad | **Memory**

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	45
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No	Assembler Message
0	Program assembled successfully

Q15. Bubble Sort – Descending Order

```
1  LDA 2500
2  DCR A
3  MOV B,A
4
5  OUTER: LXI H,2501
6  MOV C,B
7
8  INNER: MOV A,M
9  MOV D,A
10 INX H
11 MOV A,M
12 MOV E,A
13 MOV A,D
14 CMP E
15 JNC NOSWAP
16
17 MOV M,D
18 DCX H
19 MOV M,E
20 INX H
21
22 NOSWAP: DCR C
23 JNZ INNER
24 DCR B
25 JNZ OUTER
26
27 HLT
```

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	8
09C6	2502	8
09C7	2503	2
09C8	2504	1
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No	Assembler Message
0	Program assembled successfully

Q16. Bubble Sort – Ascending Order

Flag S 0 Z 1 AC 0 P 1 C 0

Load me at

```
1  LDA 2500
2  DCR A
3  MOV B,A
4
5  OUTER: LXI H,2501
6  MOV C,B
7
8  INNER: MOV A,M
9  MOV D,A
10 INX H
11 MOV A,M
12 MOV E,A
13 MOV A,D
14 CMP E
15 JC NOSWAP
16 JZ NOSWAP
17
18 MOV M,D
19 DCX H
20 MOV M,E
21 INX H
22
23 NOSWAP: DCR C
24 JNZ INNER
25 DCR B
26 JNZ OUTER
27
28 HLT
```

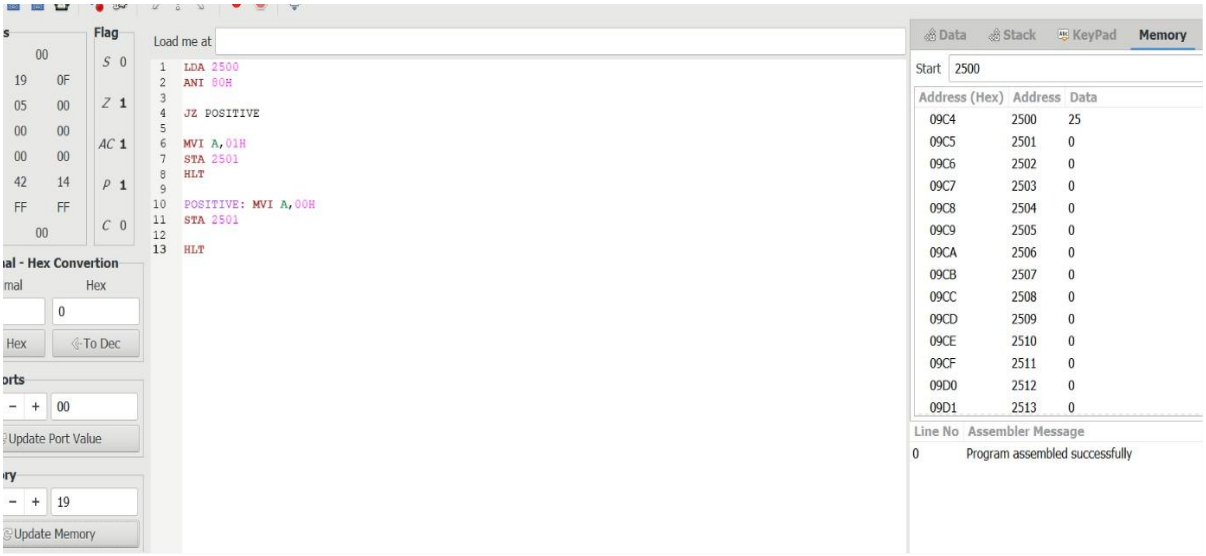
Data Stack KeyPad Memor

Start 2500

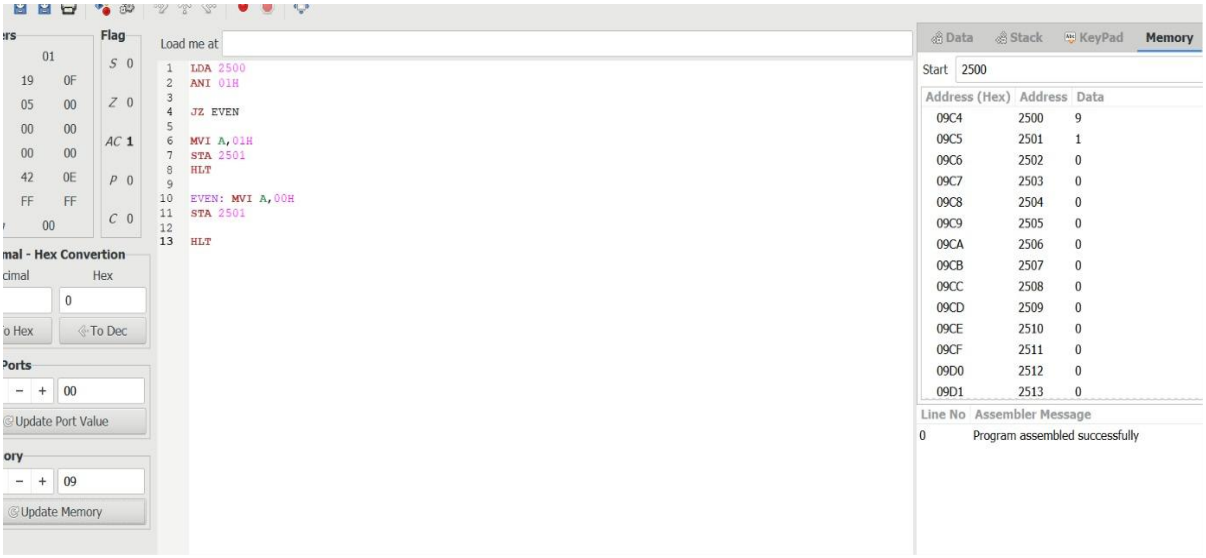
Address (Hex)	Address	Data
09C4	2500	4
09C5	2501	1
09C6	2502	2
09C7	2503	5
09C8	2504	8
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No	Assembler Message
0	Program assembled successfully

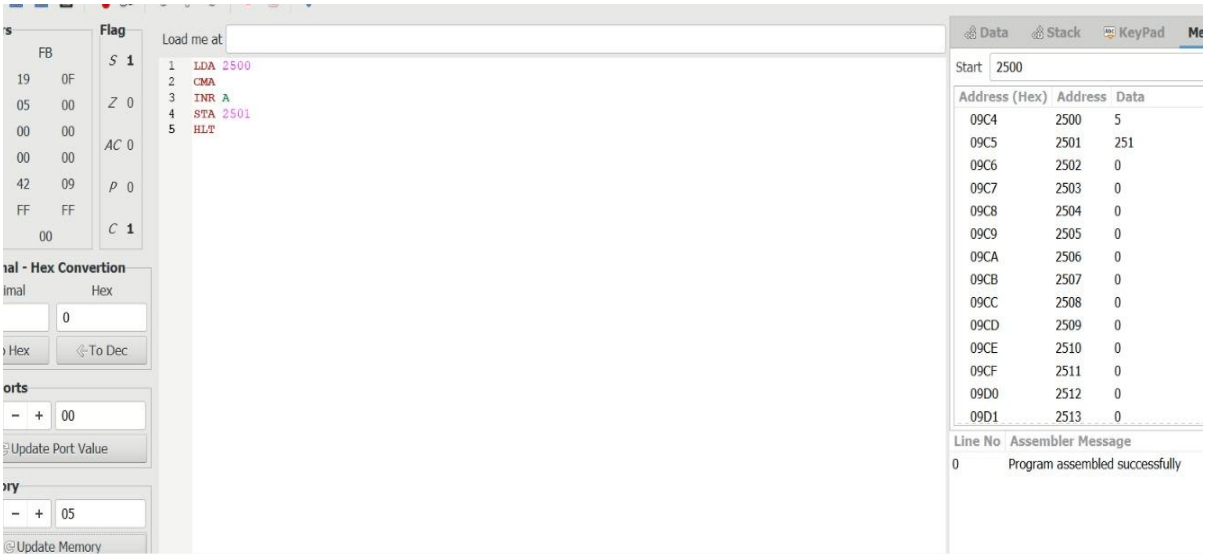
Q17. Check Whether a Number Is Positive



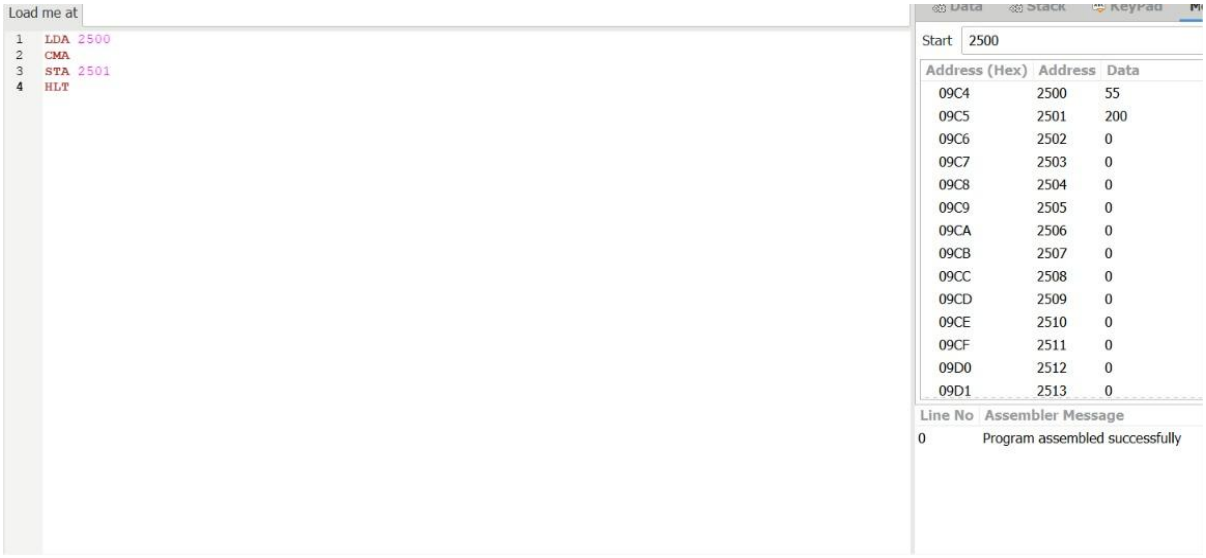
Q18. Check Whether a Number Is Odd or Even



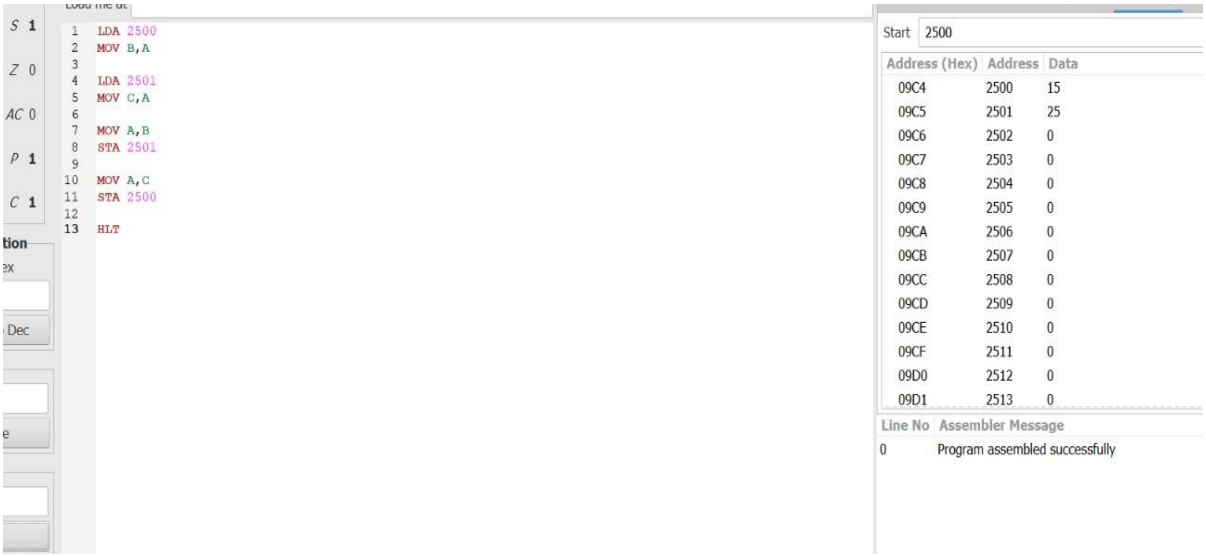
Q19. 2's Complement of a Number



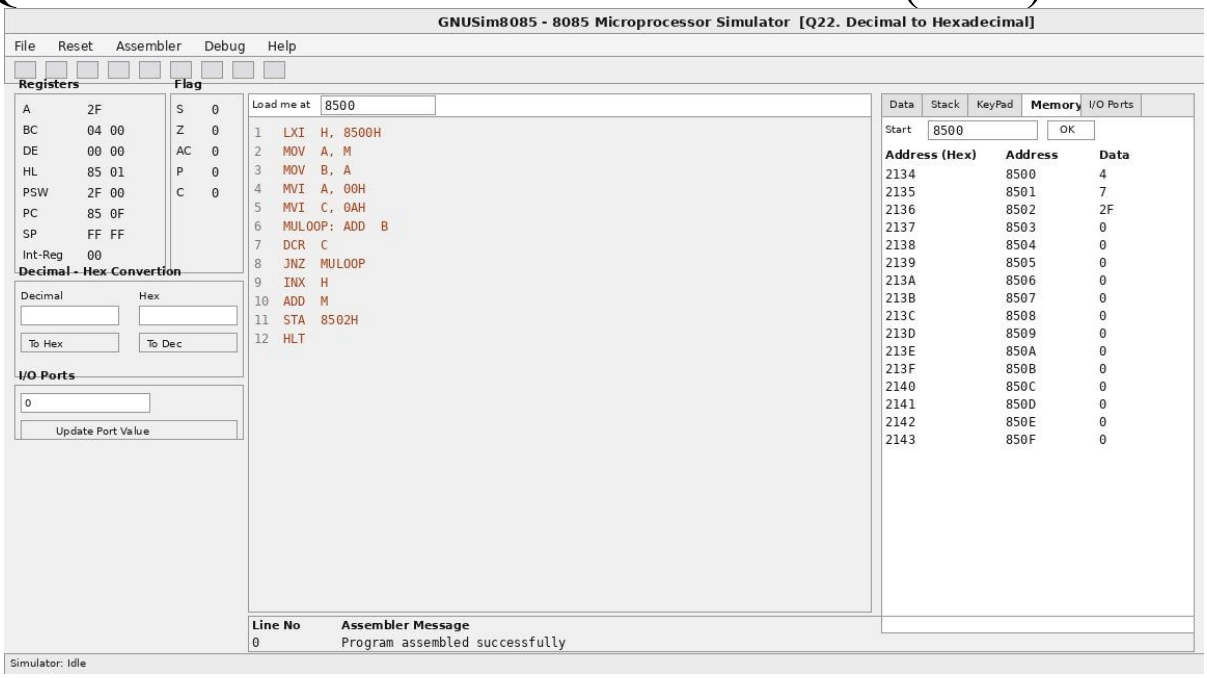
Q20. 1's Complement of a Number



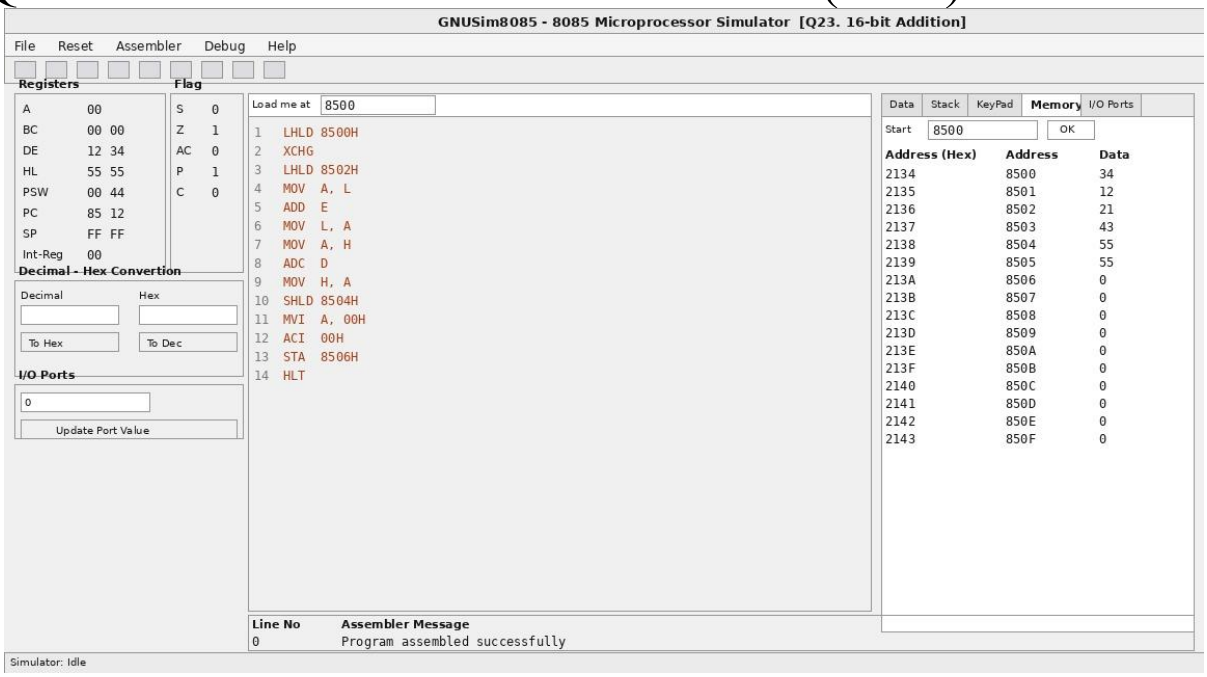
Q21. Swap Two Numbers



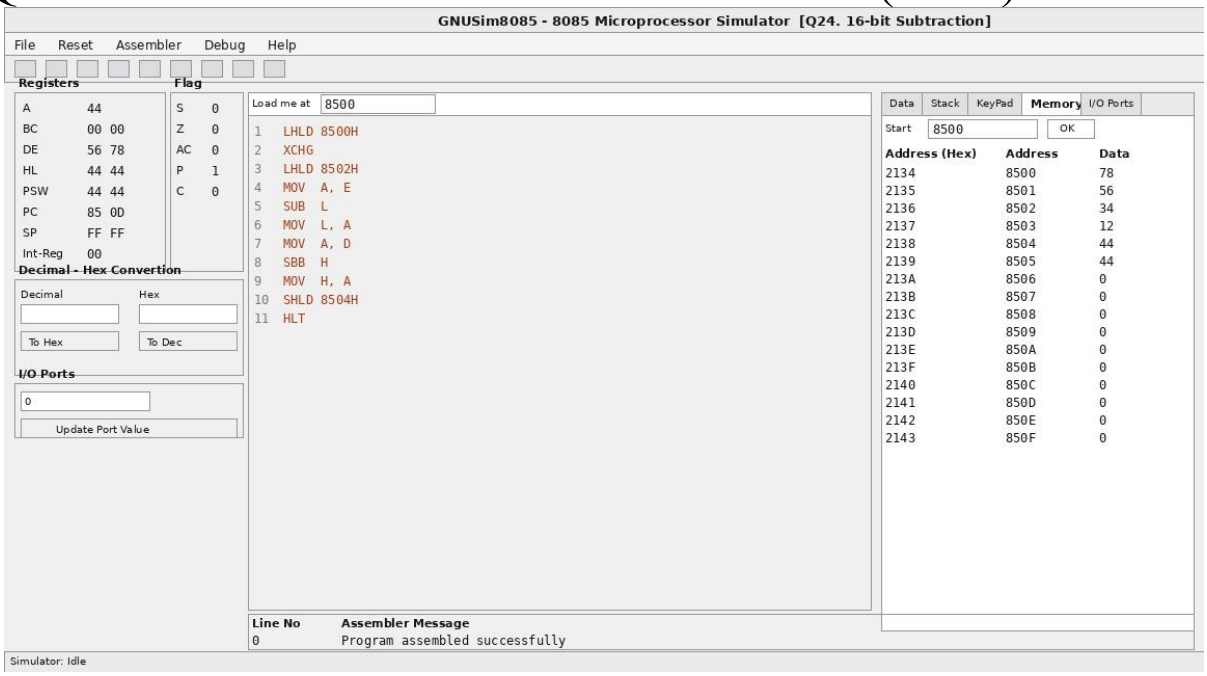
Q22. Decimal to Hexadecimal Conversion (8085)



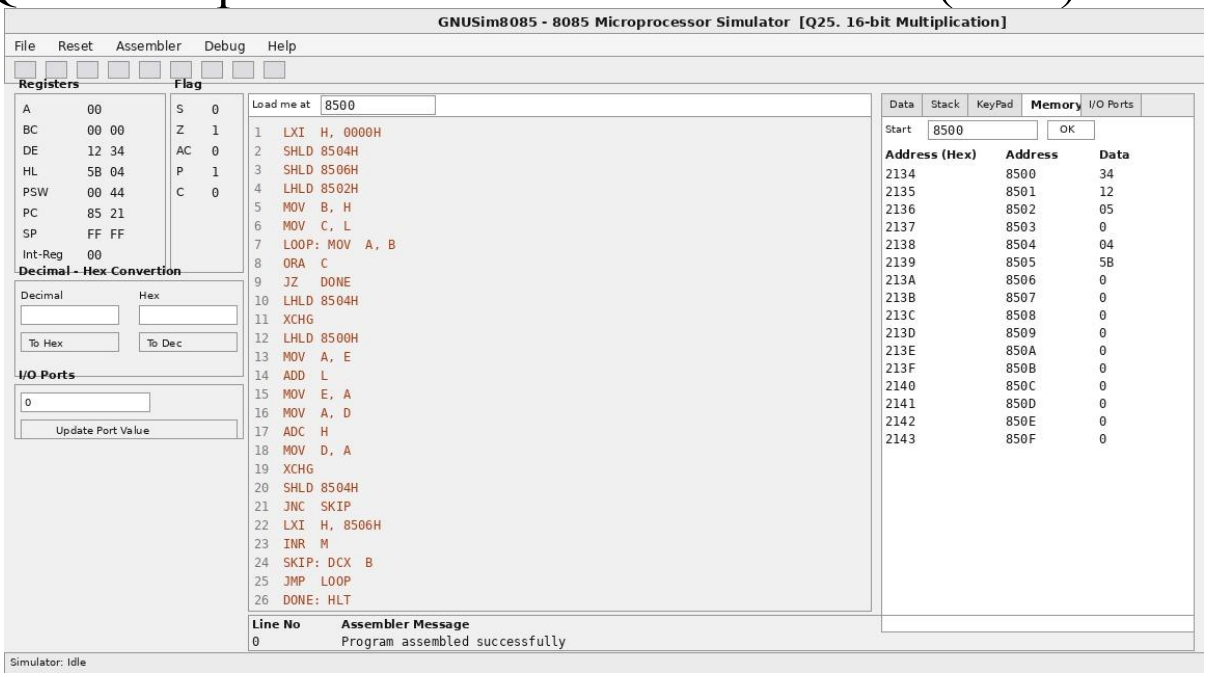
Q23. Addition of Two 16-bit Numbers (8085)



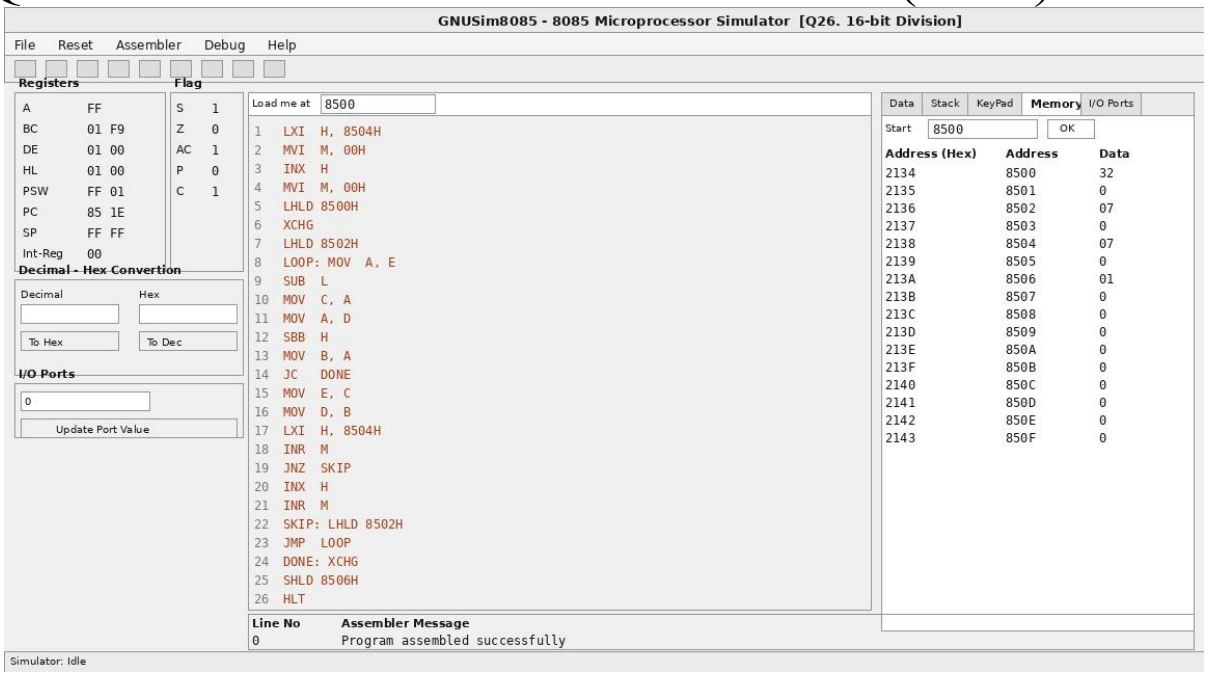
Q24. Subtraction of Two 16-bit Numbers (8085)



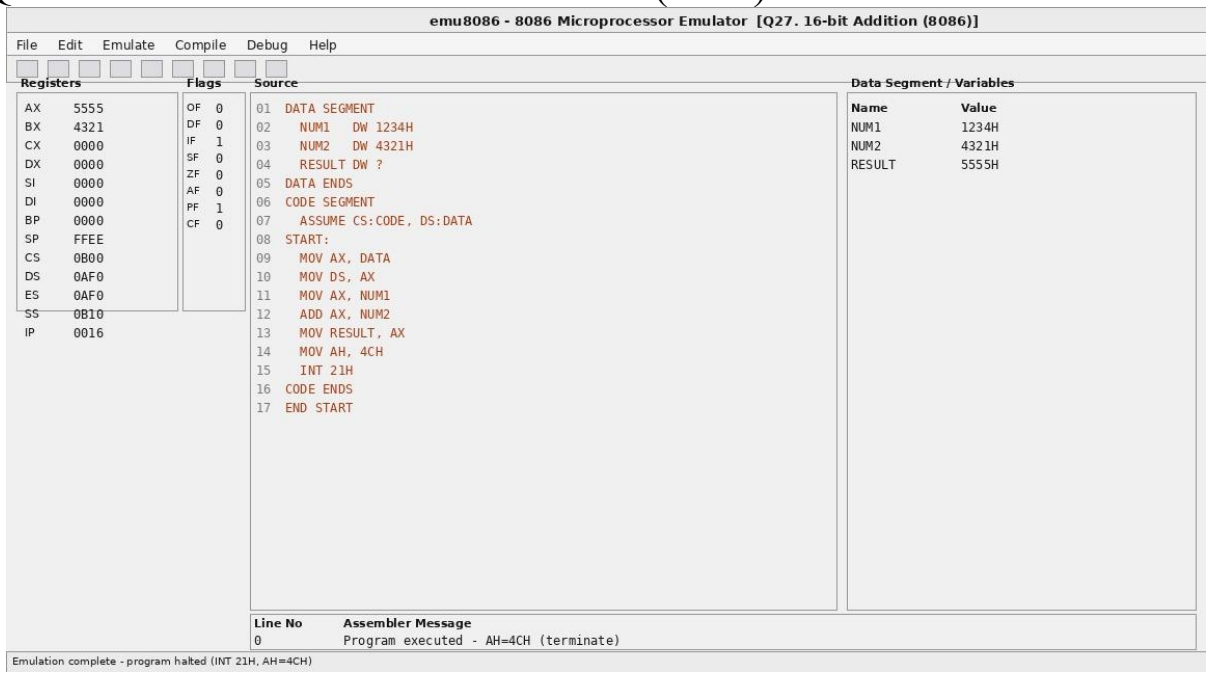
Q25. Multiplication of Two 16-bit Numbers (8085)



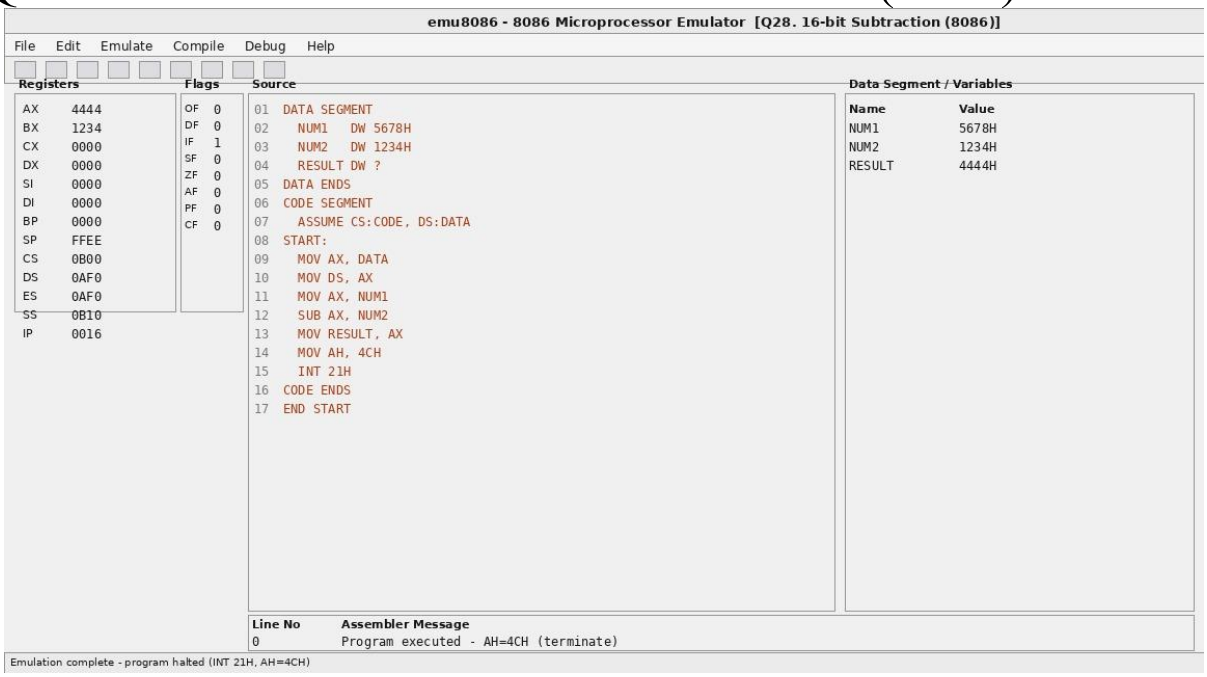
Q26. Division of Two 16-bit Numbers (8085)



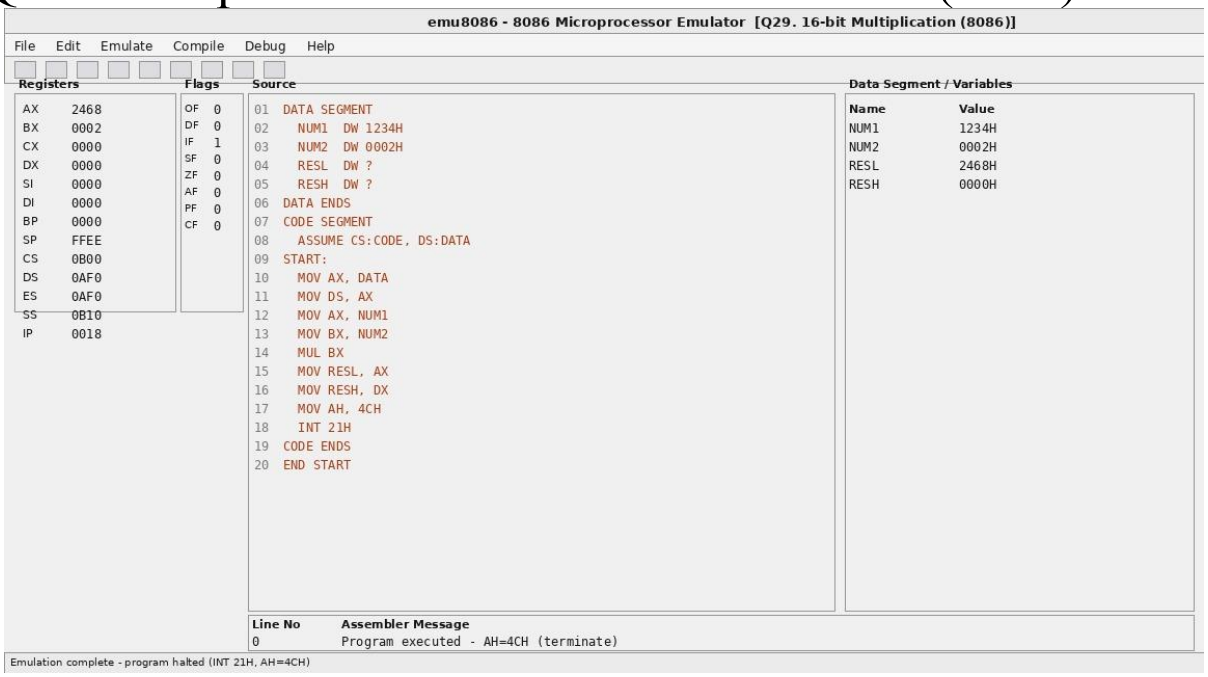
Q27. Addition of Two 16-bit Numbers (8086)



Q28. Subtraction of Two 16-bit Numbers (8086)



Q29. Multiplication of Two 16-bit Numbers (8086)



Q30. Division of Two 16-bit Numbers (8086)